CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 3: Squares and Square Roots

Every formula and result of the chapter on one sheet

1	Square	$n^2 = n \times n$
2	Last digits of a perfect square	0, 1, 4, 5, 6, 9
3	Sum of the first n odd numbers	n^2
4	Numbers between n^2 and $(n+1)^2$	$2n$
5	Consecutive squares	$(n+1)^2 - n^2 = 2n + 1$
6	Pythagorean triplet	$2m,\ m^2 - 1,\ m^2 + 1$
7	Number ending in 5	$(10a + 5)^2 = 100a(a + 1) + 25$
8	Product	$\sqrt{ab} = \sqrt{a}\sqrt{b}$
9	Quotient	$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$
10	Digits in the root	$\frac{n}{2}$ or $\frac{n+1}{2}$
11	Side of a square	$\sqrt{\text{area}}$
12	Useful roots	$\sqrt{2} \approx 1.414,\ \sqrt{3} \approx 1.732,\ \sqrt{5} \approx 2.236$